

Manaswini Ganjam

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PROFESSIONAL SUMMARY

Motivated to apply geospatial expertise and modeling tools to assess climate change impacts on forest carbon dynamics and ecosystem resilience.

EDUCATION

North Carolina State University

Ph.D. in Forestry and Environmental Resources

Raleigh, NC

Expected May 2025

Jawaharlal Nehru Technological University

M. Tech in Environmental Geomatics

Hyderabad, India

August 2015

B. Tech in Electrical and Electronics Engineering

May 2013

TECHNICAL SKILLS

- R
- PostgreSQL
- Arc GIS/ QGIS
- Python
- GEE / AWS (Geospatial cloud data)
- C, C++

EXPERIENCE

North Carolina State University

Graduate Research Assistant

Raleigh, NC

1 May 2021 – Expected May 2025

2025

- Modeled forest carbon dynamics and resilience using Biome-BGC MuSo, assessing the influences of climate change (RCP 4.5, 6.0, 8.5), stand age, and management scenarios across North American biomes.
- Calibrated and validated ecosystem process-based model simulations with flux tower data, projecting climate-driven shifts in Net Ecosystem Productivity.
- Built a geospatial database synthesizing FIA, NLCD, gSSURGO, and NASA-NEX-GDDP climate projections for the southeastern U.S., applying statistical methods to ensure data integrity.
- Created visualizations (Python, R) and briefed interdisciplinary teams on carbon flux trends, informing sustainable forest management strategies.
- Contributed to multiple research publications and presentations, showcasing ability to share research outcomes with the scientific community.
- Advanced Python, R, and PostgreSQL skills for large-scale geospatial analysis and model parameterization, adaptable to landscape-scale tools.

Central Coffee Research Institute

Project Scientist

Bengaluru, India

June 2017 – March 2018

- Performed pre-processing of high spatial resolution remote sensing data (1m).
- Conducted multiple fieldwork trips for GPS-based ground truth collection.
- Utilized object-based classification for the classification and interpretation of remote sensing data.
- Contributed to the generation of a coffee area map for India.

CSIR-National Geophysical Research Institute

Project Assistant III

Hyderabad, India

November 2015 – March 2017

- Requisitioned satellite datasets like land use, elevation, and vegetation density.
- Digitized high resolution bathymetry data.
- Conducted preprocessing of data, database creation, ancillary data collection.
- Inundation results were obtained using a numerical tsunami model.
- Conducted literature review to assign weights to the parameters using analytical hierarchy process.

- Generated vulnerability maps for Car Nicobar Islands in Indian ocean.

National Remote Sensing Centre

Hyderabad, India

Project intern

September 2014 – October 2015

- Acquired and managed multi-temporal remote sensing datasets, including Landsat and AWIFS imagery.
- Performed comprehensive data preprocessing, database creation, and ancillary data collection for geospatial analysis
- Implemented unsupervised classification techniques to assess geographical areas affected by forest fires, deforestation, and degradation.
- Conducted in-depth geospatial and statistical analyses to derive meaningful insights from remote sensing data.
- Contributed to the writing and publication of research articles, while actively collaborating on related research projects resulting in multiple peer-reviewed publications.

PUBLICATIONS

- **Ganjam, M.**, Baker, J.S., Aguilos, M., Merganičová, K. (2024). “Maximizing Net Ecosystem Productivity: Integrating Chronosequence Analysis and Ecosystem Modeling Across Contrasting Biomes in North America”. Forest Ecosystems (In Revision).
- Fuller, M. R., **Ganjam, M.**, Baker, J. S., & Abt, R. C. (2024). “Advancing forest carbon projections requires improved convergence between ecological and economic models”. Carbon Balance and Management.
- Reddy, C.S., Jha, C.S., **Manaswini, G.**, Alekhya, V.V.L.P., Vazeed Pasha, S., Satish, K.V., Diwakar, P. G., & Dadhwal V,K. “Nationwide assessment of forest burnt area in India using Resourcesat-2 AWiFS data”. Curr. Sci (2017).
- Reddy, C.S., **Manaswini, G.**, Jha, C.S., Diwakar, P.G., & Dadhwal, V.K. “Development of national database on long-term deforestation in Sri Lanka”. Journal of the Indian Society of Remote Sensing, 1-12 (2016).
- Reddy, C.S., **Manaswini, G.**, Satish, K.V., Sonali Singh., Jha, C.S., Dadhwal V,K. “Conservation priorities of forest ecosystems: Evaluation of deforestation and degradation hotspots using geospatial techniques”. Ecological Engineering volume 91 (2016):333–342.
- **Manaswini, G.**, and Reddy, C.S., “Geospatial monitoring and prioritization of forest fire incidences in Andhra Pradesh, India”. Environmental monitoring and assessment 187.10 (2015):1-12.
- Jakkula, M., Guruhappa, H., **Manaswini, G.**, & Srivastava, K. “Tsunami forces acting on ocean structures: A synthetic study”. Journal of Indian Geophysical Union, 21(6), 482-489 (2017).